2–GPU NVLink starter | 1,600 labelled contacts | 192 connected conductors

Open a GPU sheet, then connector A or B. Identical global labels join the NVLink data pins between GPUs.
Clock, power, ground, PCIe and control contacts remain labelled and unconnected for your own circuitry.

GPU0 / J1 + J2

GPU0 / V100 SXM2
A: J1 / 400 labelled contacts
B: J2 / 400 labelled contacts
192 NVLink data contacts wired

File: GPU0.kicad_sch

GPU1 / J3 + J4

GPU1 / V100 SXM2
A: J3 / 400 labelled contacts
B: J4 / 400 labelled contacts
192 NVLink data contacts wired

File: GPU1.kicad_sch

BIDIRECTIONAL GROUP PAIRINGS

GPU0 / NVHS0 <–> GPU1 / NVHS3

GPU0 / NVHS1 <–> GPU1 / NVHS2

GPU0 / NVHS2 <–> GPU1 / NVHS1

GPU0 / NVHS3 <–> GPU1 / NVHS0

GPU0 / NVHS4 <–> GPU1 / NVHS5

GPU0 / NVHS5 <–> GPU1 / NVHS4

DATA WIRING ONLY – NOT A COMPLETE WORKING NVLINK IMPLEMENTATION

External NVLink clock required on EVERY V100: 156.25 MHz differential / Connector B E20=P, F20=N.
Separate PCIe clock: 100 MHz differential / Connector A E7=P, F7=N. Clock circuits are not included.
Use suitable source/fanout outputs per GPU; also provide the required supplies, reset and supporting controls.

REFERENCE SCOPE

Full six–group two–GPU arrangement is derived from the documented pairing families.
This complete topology is not claimed as hardware–tested. Physical connector orientation and mating footprints require confirmation.
Curated from the credited community sources; see the repository sources, connection guide and standalone workbook.

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

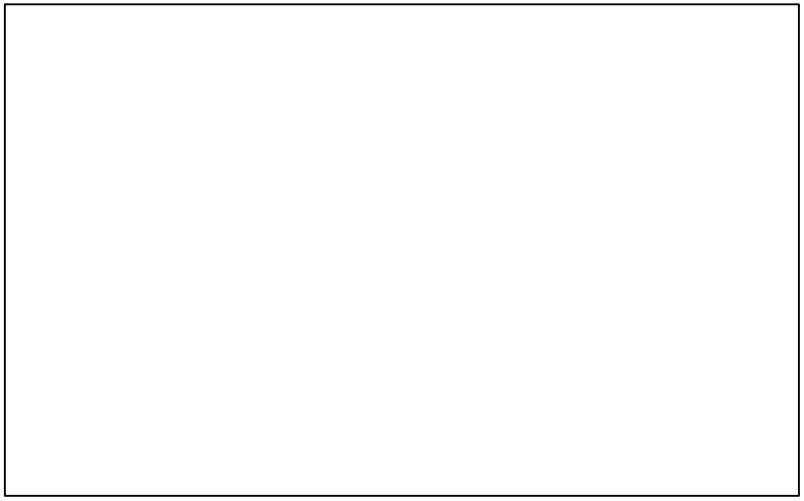
Sheet: /
File: V100_2GPU_NVLink.kicad_sch

Title: 2 V100 SXM2 – NVLink reference starter

Size: A3	Date: 2026–10–30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 1/7

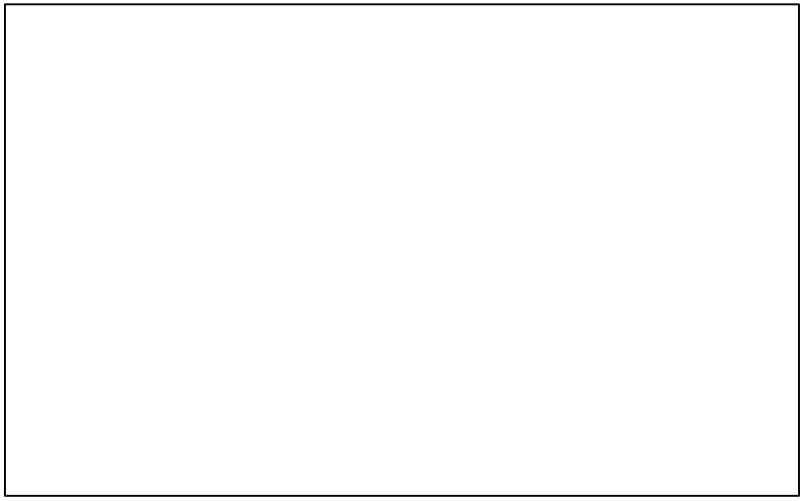
GPU0: 800 labelled contacts / 192 NVLink data contacts connected
Open Connector A for PCIe, clock and power/control contacts; open B for the NVLink net labels.

GPU0 Connector A / J1



File: GPU0_Connector_A.kicad_sch

GPU0 Connector B / J2



File: GPU0_Connector_B.kicad_sch

Supporting clock, supply and control pins remain labelled and unconnected.
Contact symbols are passive and have no assigned PCB footprint.

CLOCK CIRCUITRY REQUIRED FOR OPERATION
NVLink: 156.25 MHz differential / B:E20=P, B:F20=N
PCIe: 100 MHz differential / A:E7=P, A:F7=N

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU0 / J1 + J2/
File: GPU0.kicad_sch

Title: GPU0 – V100 SXM2 connectors

Size: A4	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 2/7

All 400 contacts retain their signal names and pad numbers. Clocks, supplies, PCIe, reset and control circuits are left for your design.

J1A

A1	GND
A2	PERp0
A3	PERn0
A4	GND
A5	PERp7
A6	PERn7
A7	GND
A8	PERp8
A9	PERn8
A10	GND
A11	PERp12
A12	PERn12
A13	GND
A14	GND
A15	UNRESOLVED/NC
A16	UNRESOLVED/NC
A17	GND
A18	PWR_GOOD
A19	GND
A20	+5V_AUX
A21	GND
A22	12V
A23	12V
A24	GND
A25	12V
A26	12V
A27	GND
A28	12V
A29	12V
A30	GND
A31	12V
A32	12V
A33	GND
A34	12V
A35	12V
A36	GND
A37	12V
A38	12V
A39	GND
A40	12V

GPU0 / A / Column A

J1B

B1	PERp1
B2	PERn1
B3	GND
B4	PERp4
B5	PERn4
B6	GND
B7	GND
B8	GND
B9	PERp9
B10	PERn9
B11	GND
B12	PERp15
B13	PERn15
B14	GND
B15	GND
B16	UNRESOLVED/NC
B17	UNRESOLVED/NC
B18	GND
B19	GND
B20	+5V_AUX
B21	GND
B22	12V
B23	12V
B24	GND
B25	12V
B26	12V
B27	GND
B28	12V
B29	12V
B30	GND
B31	12V
B32	12V
B33	GND
B34	12V
B35	12V
B36	GND
B37	12V
B38	12V
B39	GND
B40	12V

GPU0 / A / Column B

J1C

C1	GND
C2	PERp3
C3	PERn3
C4	GND
C5	PERp6
C6	PERn6
C7	GND
C8	PERp10
C9	PERn10
C10	GND
C11	PERp13
C12	PERn13
C13	GND
C14	GND
C15	UNRESOLVED/NC
C16	UNRESOLVED/NC
C17	GND
C18	I2C_ADDR0
C19	GND
C20	+5V_AUX
C21	GND
C22	12V
C23	12V
C24	GND
C25	12V
C26	12V
C27	GND
C28	12V
C29	12V
C30	GND
C31	12V
C32	12V
C33	GND
C34	12V
C35	12V
C36	GND
C37	12V
C38	12V
C39	GND
C40	12V

GPU0 / A / Column C

J1D

D1	PERp2
D2	PERn2
D3	GND
D4	PERp5
D5	PERn5
D6	GND
D7	GND
D8	GND
D9	PERp11
D10	PERn11
D11	GND
D12	PERp14
D13	PERn14
D14	GND
D15	GND
D16	UNRESOLVED/NC
D17	UNRESOLVED/NC
D18	GND
D19	GND
D20	+5V_AUX
D21	GND
D22	12V
D23	12V
D24	GND
D25	12V
D26	12V
D27	GND
D28	12V
D29	12V
D30	GND
D31	12V
D32	12V
D33	GND
D34	12V
D35	12V
D36	GND
D37	12V
D38	12V
D39	GND
D40	12V

GPU0 / A / Column D

J1E

E1	GND
E2	GND
E3	GND
E4	GND
E5	GND
E6	GND
E7	PCIE_REFCLK_P
E8	GND
E9	GND
E10	GND
E11	GND
E12	GND
E13	GND
E14	GND
E15	GND
E16	GND
E17	GND
E18	PERST#
E19	GND
E20	+5V_AUX
E21	GND
E22	12V
E23	12V
E24	GND
E25	12V
E26	12V
E27	GND
E28	12V
E29	12V
E30	GND
E31	12V
E32	12V
E33	GND
E34	12V
E35	12V
E36	GND
E37	12V
E38	12V
E39	GND
E40	12V

GPU0 / A / Column E

J1F

F1	GND
F2	GND
F3	GND
F4	GND
F5	GND
F6	GND
F7	PCIE_REFCLK_N
F8	GND
F9	GND
F10	GND
F11	GND
F12	GND
F13	GND
F14	GND
F15	GND
F16	GND
F17	GND
F18	PWR_BRAKE#
F19	GND
F20	JTAG_TDI
F21	GND
F22	12V
F23	12V
F24	GND
F25	12V
F26	12V
F27	GND
F28	12V
F29	12V
F30	GND
F31	12V
F32	12V
F33	GND
F34	12V
F35	12V
F36	GND
F37	12V
F38	12V
F39	GND
F40	12V

GPU0 / A / Column F

J1G

G1	PETp0
G2	PETn0
G3	GND
G4	PETp4
G5	PETn4
G6	GND
G7	GND
G8	GND
G9	PETp8
G10	PETn8
G11	GND
G12	PETp15
G13	PETn15
G14	GND
G15	GND
G16	UNRESOLVED/NC
G17	UNRESOLVED/NC
G18	GND
G19	GND
G20	JTAG_TDO
G21	GND
G22	12V
G23	12V
G24	GND
G25	12V
G26	12V
G27	GND
G28	12V
G29	12V
G30	GND
G31	12V
G32	12V
G33	GND
G34	12V
G35	12V
G36	GND
G37	12V
G38	12V
G39	GND
G40	12V

GPU0 / A / Column G

J1H

H1	GND
H2	PETp1
H3	PETn1
H4	GND
H5	PETp7
H6	PETn7
H7	GND
H8	PETp9
H9	PETn9
H10	GND
H11	PETp12
H12	PETn12
H13	GND
H14	GND
H15	UNRESOLVED/NC
H16	UNRESOLVED/NC
H17	GND
H18	I2C_ADDR1
H19	JTAG_SEL
H20	JTAG_TCLK
H21	GND
H22	12V
H23	12V
H24	GND
H25	12V
H26	12V
H27	GND
H28	12V
H29	12V
H30	GND
H31	12V
H32	12V
H33	GND
H34	12V
H35	12V
H36	GND
H37	12V
H38	12V
H39	GND
H40	12V

GPU0 / A / Column H

J1I

J1	PETp2
J2	PETn2
J3	GND
J4	PETp5
J5	PETn5
J6	GND
J7	GND
J8	GND
J9	PETp11
J10	PETn11
J11	GND
J12	PETp14
J13	PETn14
J14	GND
J15	GND
J16	UNRESOLVED/NC
J17	UNRESOLVED/NC
J18	GND
J19	GND
J20	JTAG_TMS
J21	GND
J22	12V
J23	12V
J24	GND
J25	12V
J26	12V
J27	GND
J28	12V
J29	12V
J30	GND
J31	12V
J32	12V
J33	GND
J34	12V
J35	12V
J36	GND
J37	12V
J38	12V
J39	GND
J40	12V

GPU0 / A / Column J

J1J

K1	GND
K2	PETp3
K3	PETn3
K4	GND
K5	PETp6
K6	PETn6
K7	GND
K8	PETp10
K9	PETn10
K10	GND
K11	PETp13
K12	PETn13
K13	GND
K14	GND
K15	UNRESOLVED/NC
K16	UNRESOLVED/NC
K17	GND
K18	PWR_EN
K19	THERM_OVERT#
K20	JTAG_TRST#
K21	GND
K22	12V
K23	12V
K24	GND
K25	12V
K26	12V
K27	GND
K28	12V
K29	12V
K30	GND
K31	12V
K32	12V
K33	GND
K34	12V
K35	12V
K36	GND
K37	12V
K38	12V
K39	GND
K40	12V

GPU0 / A / Column K

Separate PCIe reference clock: 100 MHz differential on A/E7 (P) and A/F7 (N). Clock circuitry is not included.

A/B7 and A/A39: disputed GND labels (reference presence loop, PRSNT2_A/B).
Ground both only if your carrier omits presence detection and its controller does not depend on it.
For presence detection, implement and validate the loop. Grounding its readout can force a false present state.
Source: Quanta Big Basin sheets 15–16. See docs/presence-detection.md.

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

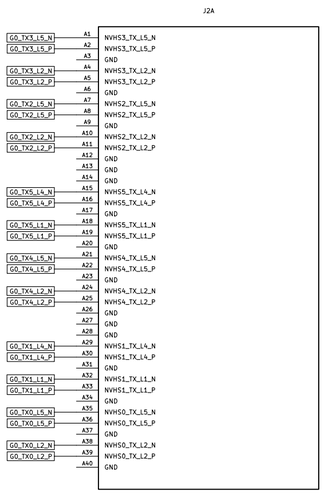
Sheet: /GPU0 / J1 + J2/GPU0 Connector A / J1/
File: GPU0_Connector_A.kicad_sch

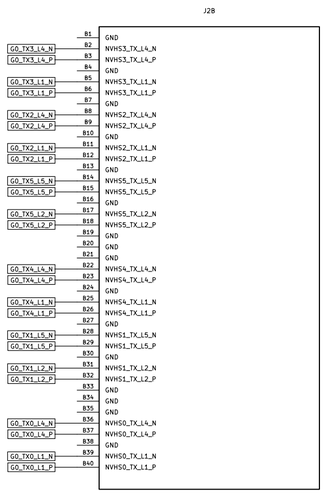
Title: GPU0 — Connector A — 400 contacts

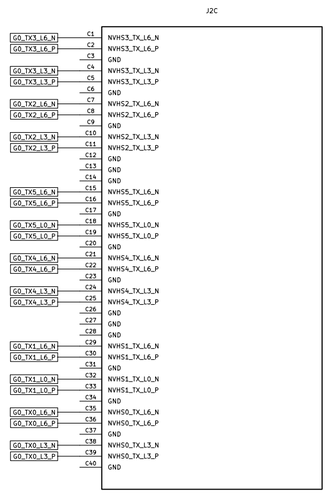
Size: A3	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 3/7

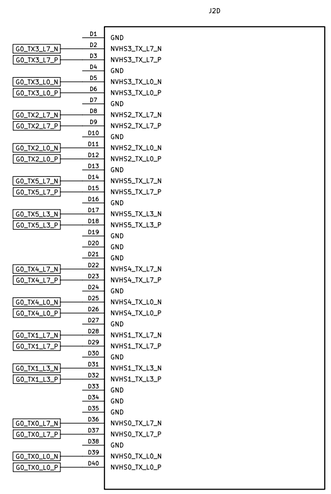
NVLink data pins connect through matching global labels. Clock, ground, power and control contacts keep their names and remain unconnected.

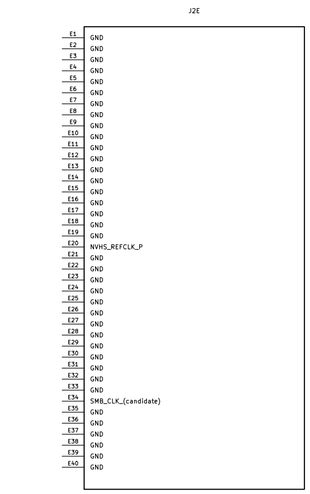
DATA WIRING ONLY — EXTERNAL CLOCK REQUIRED
Each V100: 156.25 MHz differential NVHS clock on B/E20 (P) and B/F20 (N). Clock circuitry is not included.

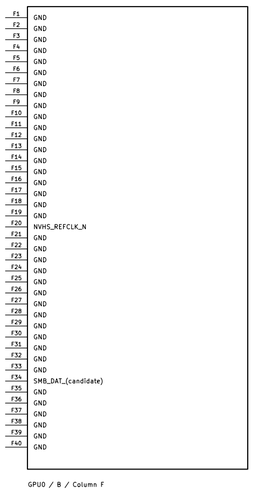


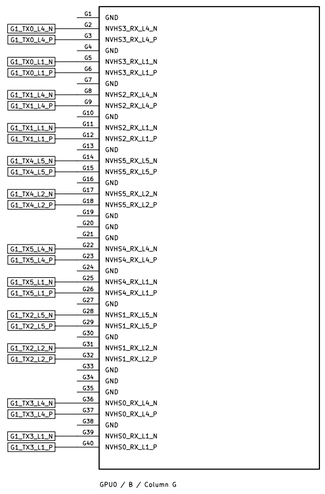


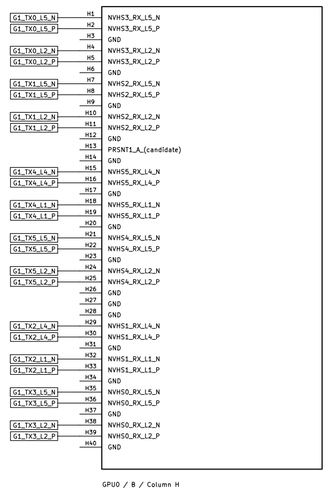


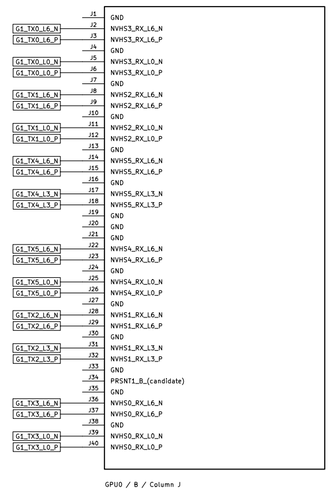


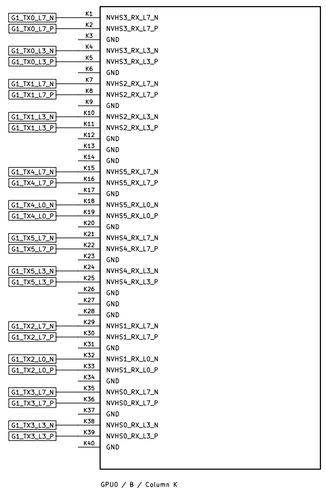












NET LABEL KEY / Example: G0_TX0_L0_P

G0
Transmitting GPU0

TX0
NVHS0 transmit

L0
Lane 0

P / N
Positive / negative

Both ends use the transmitting contact name: full signal names remain beside every pin.
Only NVLink data is wired. An unconnected contact is not an official NVIDIA NC assignment.

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

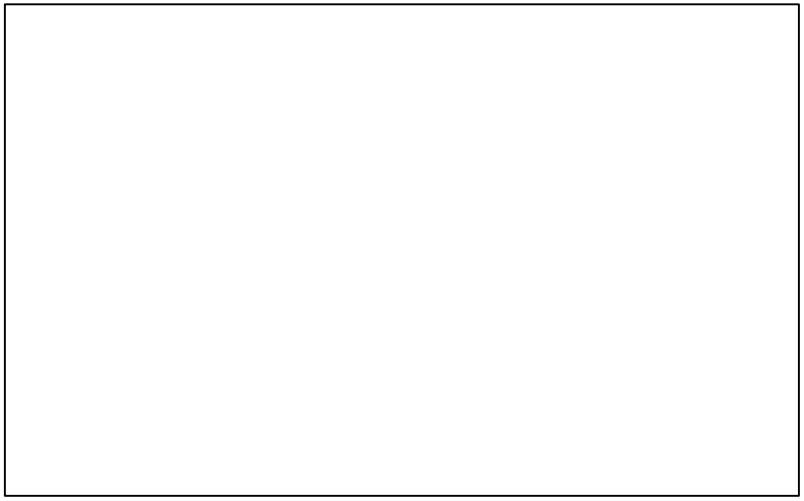
Sheet: /GPU0 / J1 + J2/GPU0 Connector B / J2/
File: GPU0_Connector_B.kicad_sch

Title: GPU0 — Connector B — 400 contacts

Size: A2	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 4/7

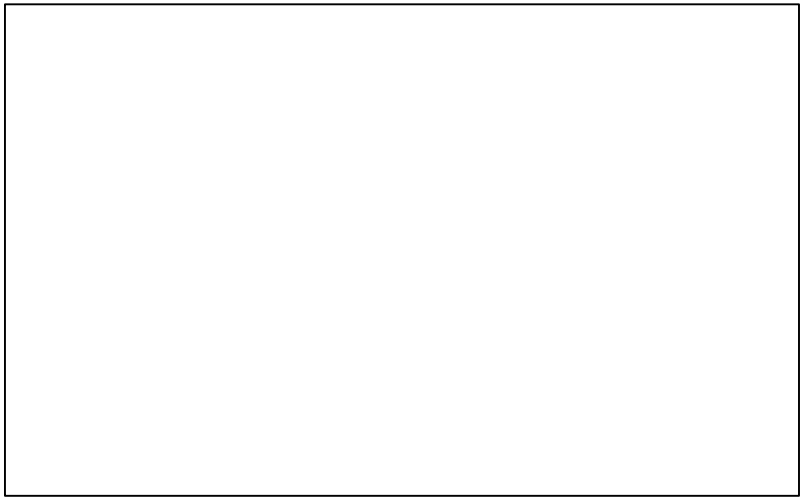
GPU1: 800 labelled contacts / 192 NVLink data contacts connected
Open Connector A for PCIe, clock and power/control contacts; open B for the NVLink net labels.

GPU1 Connector A / J3



File: GPU1_Connector_A.kicad_sch

GPU1 Connector B / J4



File: GPU1_Connector_B.kicad_sch

Supporting clock, supply and control pins remain labelled and unconnected.
Contact symbols are passive and have no assigned PCB footprint.

CLOCK CIRCUITRY REQUIRED FOR OPERATION
NVLink: 156.25 MHz differential / B:E20=P, B:F20=N
PCIe: 100 MHz differential / A:E7=P, A:F7=N

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU1 / J3 + J4/
File: GPU1.kicad_sch

Title: GPU1 – V100 SXM2 connectors

Size: A4	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 5/7

All 400 contacts retain their signal names and pad numbers. Clocks, supplies, PCIe, reset and control circuits are left for your design.

J3A

A1	GND
A2	PERp0
A3	PERn0
A4	GND
A5	PERp7
A6	PERn7
A7	GND
A8	PERp8
A9	PERn8
A10	GND
A11	PERp12
A12	PERn12
A13	GND
A14	GND
A15	UNRESOLVED/NC
A16	UNRESOLVED/NC
A17	GND
A18	PWR_GOOD
A19	GND
A20	+5V_AUX
A21	GND
A22	12V
A23	12V
A24	GND
A25	12V
A26	12V
A27	GND
A28	12V
A29	12V
A30	GND
A31	12V
A32	12V
A33	GND
A34	12V
A35	12V
A36	GND
A37	12V
A38	12V
A39	GND
A40	12V

GPU1 / A / Column A

J3B

B1	PERp1
B2	PERn1
B3	GND
B4	PERp4
B5	PERn4
B6	GND
B7	GND
B8	GND
B9	PERp9
B10	PERn9
B11	GND
B12	PERp15
B13	PERn15
B14	GND
B15	GND
B16	UNRESOLVED/NC
B17	UNRESOLVED/NC
B18	GND
B19	GND
B20	+5V_AUX
B21	GND
B22	12V
B23	12V
B24	GND
B25	12V
B26	12V
B27	GND
B28	12V
B29	12V
B30	GND
B31	12V
B32	12V
B33	GND
B34	12V
B35	12V
B36	GND
B37	12V
B38	12V
B39	GND
B40	12V

GPU1 / A / Column B

J3C

C1	GND
C2	PERp3
C3	PERn3
C4	GND
C5	PERp6
C6	PERn6
C7	GND
C8	PERp10
C9	PERn10
C10	GND
C11	PERp13
C12	PERn13
C13	GND
C14	GND
C15	UNRESOLVED/NC
C16	UNRESOLVED/NC
C17	GND
C18	I2C_ADDR0
C19	GND
C20	+5V_AUX
C21	GND
C22	12V
C23	12V
C24	GND
C25	12V
C26	12V
C27	GND
C28	12V
C29	12V
C30	GND
C31	12V
C32	12V
C33	GND
C34	12V
C35	12V
C36	GND
C37	12V
C38	12V
C39	GND
C40	12V

GPU1 / A / Column C

J3D

D1	PERp2
D2	PERn2
D3	GND
D4	PERp5
D5	PERn5
D6	GND
D7	GND
D8	GND
D9	PERp11
D10	PERn11
D11	GND
D12	PERp14
D13	PERn14
D14	GND
D15	GND
D16	UNRESOLVED/NC
D17	UNRESOLVED/NC
D18	GND
D19	GND
D20	+5V_AUX
D21	GND
D22	12V
D23	12V
D24	GND
D25	12V
D26	12V
D27	GND
D28	12V
D29	12V
D30	GND
D31	12V
D32	12V
D33	GND
D34	12V
D35	12V
D36	GND
D37	12V
D38	12V
D39	GND
D40	12V

GPU1 / A / Column D

J3E

E1	GND
E2	GND
E3	GND
E4	GND
E5	GND
E6	GND
E7	PCIE_REFCLK_P
E8	GND
E9	GND
E10	GND
E11	GND
E12	GND
E13	GND
E14	GND
E15	GND
E16	GND
E17	GND
E18	PERST#
E19	GND
E20	+5V_AUX
E21	GND
E22	12V
E23	12V
E24	GND
E25	12V
E26	12V
E27	GND
E28	12V
E29	12V
E30	GND
E31	12V
E32	12V
E33	GND
E34	12V
E35	12V
E36	GND
E37	12V
E38	12V
E39	GND
E40	12V

GPU1 / A / Column E

J3F

F1	GND
F2	GND
F3	GND
F4	GND
F5	GND
F6	GND
F7	PCIE_REFCLK_N
F8	GND
F9	GND
F10	GND
F11	GND
F12	GND
F13	GND
F14	GND
F15	GND
F16	GND
F17	GND
F18	PWR_BRAKE#
F19	GND
F20	JTAG_TDI
F21	GND
F22	12V
F23	12V
F24	GND
F25	12V
F26	12V
F27	GND
F28	12V
F29	12V
F30	GND
F31	12V
F32	12V
F33	GND
F34	12V
F35	12V
F36	GND
F37	12V
F38	12V
F39	GND
F40	12V

GPU1 / A / Column F

J3G

G1	PETp0
G2	PETn0
G3	GND
G4	PETp4
G5	PETn4
G6	GND
G7	GND
G8	GND
G9	PETp8
G10	PETn8
G11	GND
G12	PETp15
G13	PETn15
G14	GND
G15	GND
G16	UNRESOLVED/NC
G17	UNRESOLVED/NC
G18	GND
G19	GND
G20	JTAG_TDO
G21	GND
G22	12V
G23	12V
G24	GND
G25	12V
G26	12V
G27	GND
G28	12V
G29	12V
G30	GND
G31	12V
G32	12V
G33	GND
G34	12V
G35	12V
G36	GND
G37	12V
G38	12V
G39	GND
G40	12V

GPU1 / A / Column G

J3H

H1	GND
H2	PETp1
H3	PETn1
H4	GND
H5	PETp7
H6	PETn7
H7	GND
H8	PETp9
H9	PETn9
H10	GND
H11	PETp12
H12	PETn12
H13	GND
H14	GND
H15	UNRESOLVED/NC
H16	UNRESOLVED/NC
H17	GND
H18	I2C_ADDR1
H19	JTAG_SEL
H20	JTAG_TCLK
H21	GND
H22	12V
H23	12V
H24	GND
H25	12V
H26	12V
H27	GND
H28	12V
H29	12V
H30	GND
H31	12V
H32	12V
H33	GND
H34	12V
H35	12V
H36	GND
H37	12V
H38	12V
H39	GND
H40	12V

GPU1 / A / Column H

J3I

J1	PETp2
J2	PETn2
J3	GND
J4	PETp5
J5	PETn5
J6	GND
J7	GND
J8	GND
J9	PETp11
J10	PETn11
J11	GND
J12	PETp14
J13	PETn14
J14	GND
J15	GND
J16	UNRESOLVED/NC
J17	UNRESOLVED/NC
J18	GND
J19	GND
J20	JTAG_TMS
J21	GND
J22	12V
J23	12V
J24	GND
J25	12V
J26	12V
J27	GND
J28	12V
J29	12V
J30	GND
J31	12V
J32	12V
J33	GND
J34	12V
J35	12V
J36	GND
J37	12V
J38	12V
J39	GND
J40	12V

GPU1 / A / Column J

J3J

K1	GND
K2	PETp3
K3	PETn3
K4	GND
K5	PETp6
K6	PETn6
K7	GND
K8	PETp10
K9	PETn10
K10	GND
K11	PETp13
K12	PETn13
K13	GND
K14	GND
K15	UNRESOLVED/NC
K16	UNRESOLVED/NC
K17	GND
K18	PWR_EN
K19	THERM_OVERT#
K20	JTAG_TRST#
K21	GND
K22	12V
K23	12V
K24	GND
K25	12V
K26	12V
K27	GND
K28	12V
K29	12V
K30	GND
K31	12V
K32	12V
K33	GND
K34	12V
K35	12V
K36	GND
K37	12V
K38	12V
K39	GND
K40	12V

GPU1 / A / Column K

Separate PCIe reference clock: 100 MHz differential on A/E7 (P) and A/F7 (N). Clock circuitry is not included.

A/B7 and A/A39: disputed GND labels (reference presence loop, PRSNT2_A/B).
Ground both only if your carrier omits presence detection and its controller does not depend on it.
For presence detection, implement and validate the loop. Grounding its readout can force a false present state.
Source: Quanta Big Basin sheets 15–16. See docs/presence-detection.md.

Logical NVLink reference; no hardware qualification or footprint assignment

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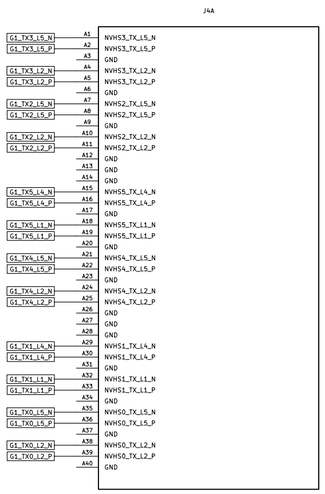
Sheet: /GPU1 / J3 + J4/GPU1 Connector A / J3/
File: GPU1_Connector_A.kicad_sch

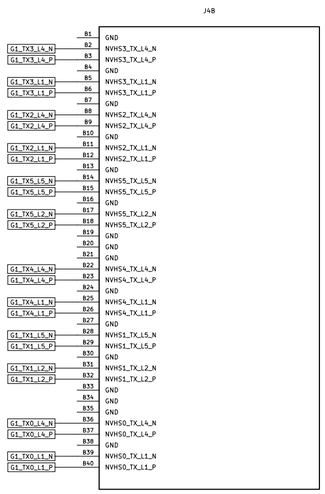
Title: GPU1 — Connector A — 400 contacts

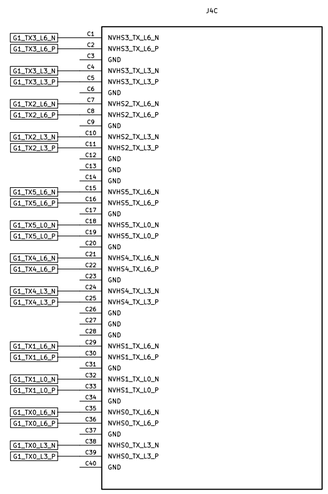
Size: A3	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 6/7

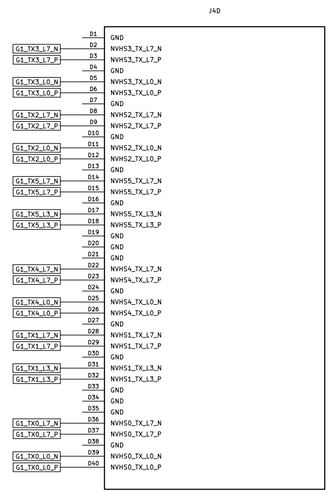
NVLink data pins connect through matching global labels. Clock, ground, power and control contacts keep their names and remain unconnected.

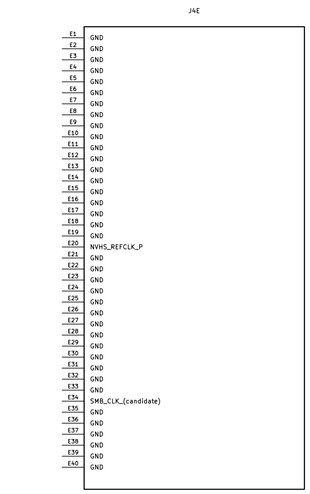
DATA WIRING ONLY — EXTERNAL CLOCK REQUIRED
Each V100: 156.25 MHz differential NVHS clock on B/E20 (P) and B/F20 (N). Clock circuitry is not included.

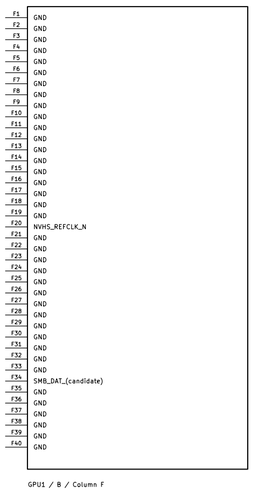


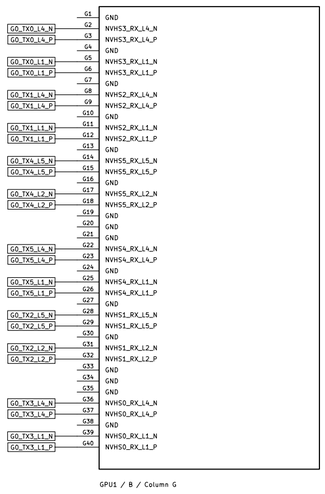


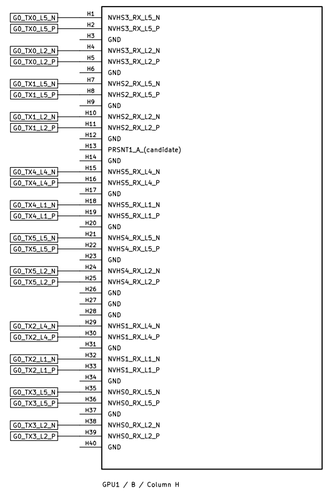


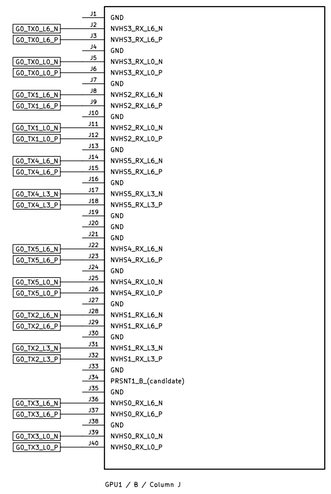


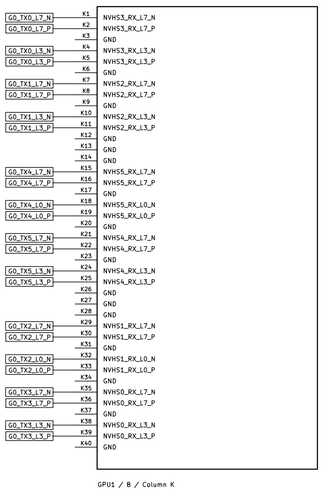












NET LABEL KEY / Example: G0_TX0_L0_P

G0
Transmitting GPU0

TX0
NVHS0 transmit

L0
Lane 0

P / N
Positive / negative

Both ends use the transmitting contact name: full signal names remain beside every pin.
Only NVLink data is wired. An unconnected contact is not an official NVIDIA NC assignment.

Logical NVLink reference: no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU1 / J3 + J4/GPU1 Connector B / J4/
File: GPU1_Connector_B.kicad_sch

Title: GPU1 — Connector B — 400 contacts

Size: A2	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 7/7